

MATH1231 CALCULUS

Chapter 1B

Term II 2026

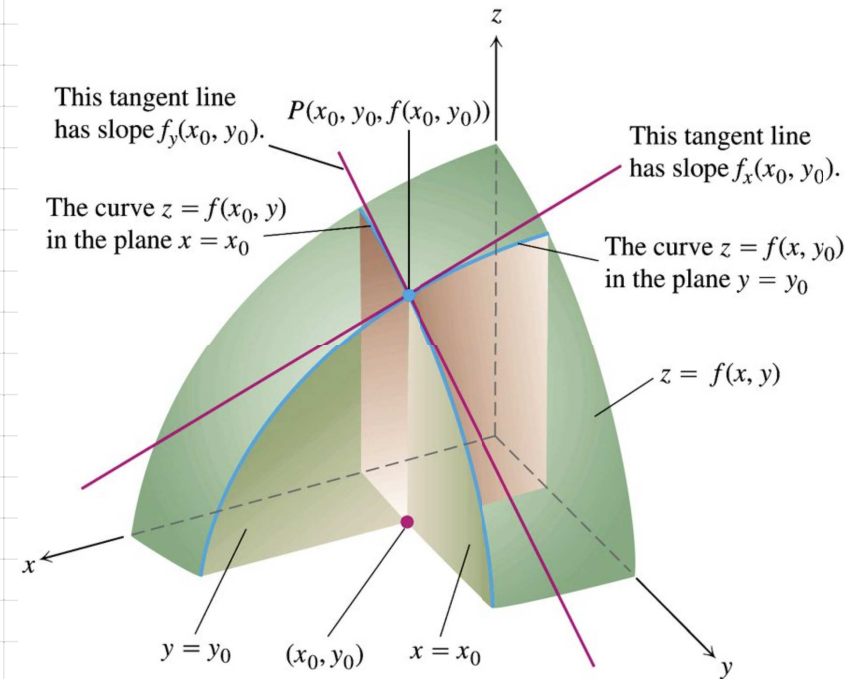
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Tangent Planes and Surface Normals

Consider the 1d tangent line first:



$\vec{u} = (1, 0, f_x(x_0, y_0))$ is tangent to cut through f with $y = y_0$ at $x = x_0$

$\vec{v} = (0, 1, f_y(x_0, y_0))$ is tangent to cut through f with $x = x_0$ at $y = y_0$

vectors $\vec{u}$ and $\vec{v}$ span the tangent plane touching f at (x_0, y_0) .

- **normal vector** to the tangent plane is $\vec{n} = \vec{u} \times \vec{v}$: The cross product of $\vec{u}$ and $\vec{v}$ is defined by

$$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & f_x \\ 0 & 1 & f_y \end{vmatrix} \quad (1)$$

$$= \hat{i}(0 - f_x) - \hat{j}(f_y - 0) + \hat{k}(1 - 0) \quad (2)$$

$$= (-f_x, -f_y, 1) \quad (3)$$

Note that $-\vec{n}$ is also normal to the tangent plane. Now that we have a normal vector, we can easily construct the tangent plane:

Constructing the tangent plane:

Tangent Planes and Surface Normals

Suppose (x_0, y_0, z_0) is a point on the surface $z = f(x, y)$.

- The **tangent plane** through (x_0, y_0, z_0) has equation

$$z = z_0 + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

- A **normal vector** to the surface at (x_0, y_0, z_0) is

$$\begin{pmatrix} f_x(x_0, y_0) \\ f_y(x_0, y_0) \\ -1 \end{pmatrix}.$$

Example

Find the upward pointing normal vector and the tangent plane to the surface $z = f(x, y) = 9 - x^2 - y^2$ at $(x_0, y_0) = (1, 2)$.

(1.4) Total Differential Approximation

If $f = f(x)$ then near a given point x_0

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0)$$

$$\Delta f = f(x) - f(x_0)$$

- ▶ The differential approximation to the difference is

$$\Delta f \approx f'(x_0)(x - x_0)$$

- ▶ Geometrically this approximates points on the curve $y = f(x)$ near x_0 by points on the tangent line near this point.

If $F = F(x, y)$ then near a given point (x_0, y_0)

$$F(x, y) \approx F(x_0, y_0) + F_x(x_0, y_0)(x - x_0) + F_y(x_0, y_0)(y - y_0)$$

$$\Delta F = F(x, y) - F(x_0, y_0)$$

Writing $\Delta x \equiv x - x_0$ and $\Delta y \equiv y - y_0$,
the total differential approximation to the difference ΔF is

$$\Delta F \approx F_x(x_0, y_0)\Delta x + F_y(x_0, y_0)\Delta y$$

- ▶ Geometrically this approximates points on the surface $z = F(x, y)$ near (x_0, y_0) by points on the tangent plane near this point.
- ▶ Note that in the limit of infinitesimal changes $\Delta x \rightarrow dx$ and $\Delta y \rightarrow dy$, we obtain the exact total differential of F :

$$dF = \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy$$

and hence the name “total differential” approximation.

Uses of the total differential approximation

- ▶ To estimate changes in output given changes in input
- ▶ To estimate upper bounds on errors

$$\begin{aligned}\text{Error} &\approx |\Delta F| \\ &= \left| \frac{\partial F}{\partial x} \Delta x + \frac{\partial F}{\partial y} \Delta y \right| \\ &\leq \left| \frac{\partial F}{\partial x} \right| |\Delta x| + \left| \frac{\partial F}{\partial y} \right| |\Delta y|\end{aligned}$$

Example

The resonance frequency of an LC circuit is given by $f(x, y) = \frac{1}{2\pi\sqrt{xy}}$,

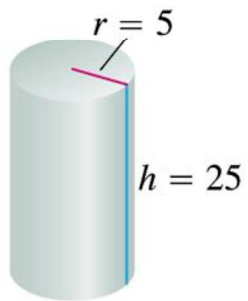
where x is the inductance and y is the capacitance.

If x is decreased by 1.5% and y is decreased by 0.5%, find the approximate percentage change in f .

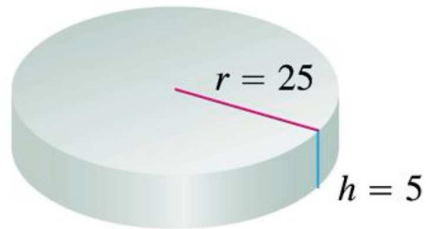
Example

The dimensions of a cylinder are measured to the nearest millimeter using a measuring tape. The radius is measured to be 5 cm and the height is measured to be 12 cm. What should we expect the maximum percentage error in calculating the volume V to be?

Applications matter! A manufacturer produces cylindrical storage tanks with height 25m and radius 5m. As a quality control engineer hired by the company, perform an analysis on how sensitive the tanks' volumes are to small variations in height and radius. Which measurement (height or radius) would you advise the company to pay particular attention to?



(a)

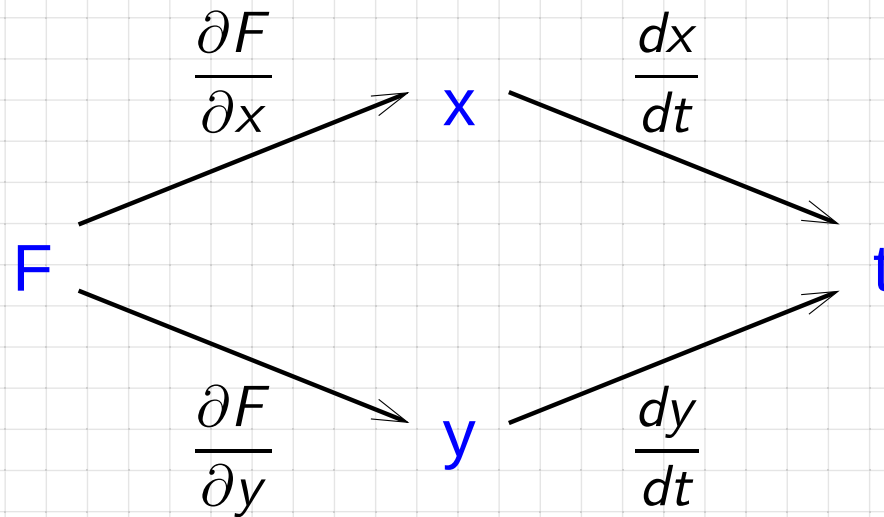


(b)

(1.5) Chain Rules

If $F = F(x, y)$ and $x = x(t)$, $y = y(t)$ then

$$\frac{dF}{dt} = \frac{\partial F}{\partial x} \frac{dx}{dt} + \frac{\partial F}{\partial y} \frac{dy}{dt}$$



To compute $\frac{dF}{dt}$ sum the paths (left to right) from F to t and multiply derivatives on a path.

Justification

A small change Δt in t produces small changes Δx in x and Δy in y which produces a small change ΔF in F – use differential approximations.

$$\Delta F \approx \frac{\partial F}{\partial x} \Delta x + \frac{\partial F}{\partial y} \Delta y$$
$$\frac{\Delta F}{\Delta t} \approx \frac{\partial F}{\partial x} \frac{\Delta x}{\Delta t} + \frac{\partial F}{\partial y} \frac{\Delta y}{\Delta t}$$

$$\frac{\Delta F}{\Delta t} = \frac{F(t + \Delta t) - F(t)}{\Delta t} \rightarrow \frac{dF}{dt}$$
$$\frac{\Delta x}{\Delta t} = \frac{x(t + \Delta t) - x(t)}{\Delta t} \rightarrow \frac{dx}{dt}$$
$$\frac{\Delta y}{\Delta t} = \frac{y(t + \Delta t) - y(t)}{\Delta t} \rightarrow \frac{dy}{dt}$$

Note: Never treat a partial derivative as a quotient.

Example

Given $z = F(x, y) = x^2y + 3xy^4$ and $x = e^t$, $y = \sin t$, use the chain rule to find $\frac{dz}{dt}$.

Example

A particle moves on the path defined by the parametric equations $x = t$, $y = t^3 + 2$. The temperature at a point (x, y) on the path is given by $T = x^2 + y^2$. Find the rate of change of temperature T on the path.

Example

The ideal gas law for the pressure p of n moles of gas in a volume V at absolute temperature T is $p = nRT/V$, where $R = 8.31 \text{ N m mol}^{-1} \text{ K}^{-1}$ is the gas constant.

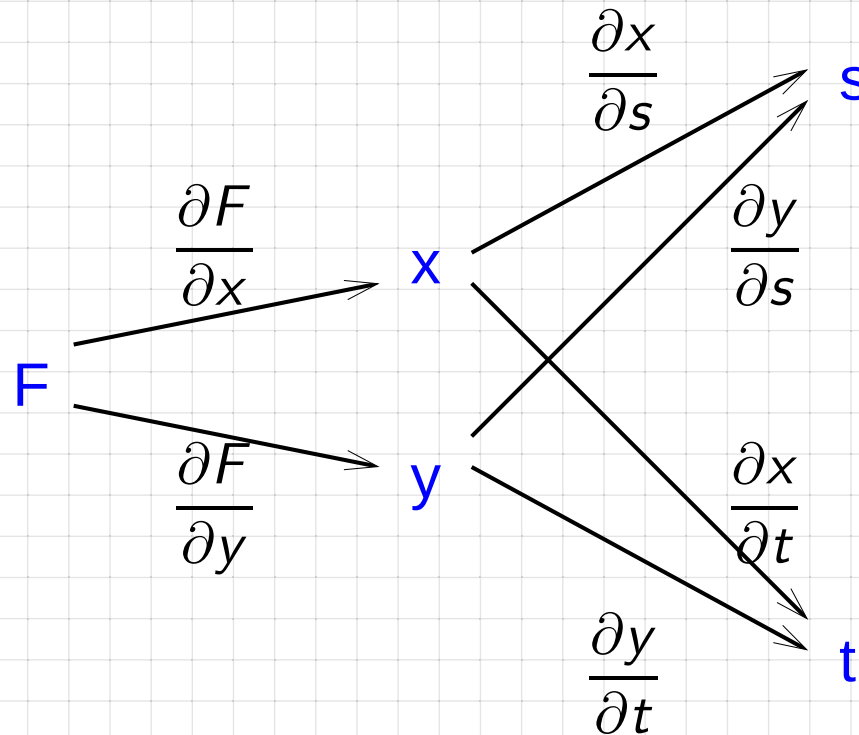
Given 1 mol of gas at $T = 300 \text{ K}$ in a volume $V = 100 \text{ m}^3$, calculate the rate of change of its pressure at the instant when V and T are changing at the rates $dV/dt = 0.2 \text{ m}^3/\text{s}$ and $dT/dt = 0.1 \text{ K/s}$.

Chain Rule

If $F = F(x, y)$ and $x = x(s, t), y = y(s, t)$ then

$$\frac{\partial F}{\partial t} = \frac{\partial F}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial F}{\partial y} \frac{\partial y}{\partial t}$$

$$\frac{\partial F}{\partial s} = \frac{\partial F}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial F}{\partial y} \frac{\partial y}{\partial s}$$



To compute $\frac{\partial F}{\partial s}$ sum the paths (left to right) from F to s and multiply partial derivatives on a path.

Example

Given $z = e^x \sin(y)$ and $x = st^2$, $y = s^2t$, use the chain rule to find $\frac{\partial z}{\partial s}$.

Example

Let f have continuous partial derivatives. Show that

$$z = f(u - v, v - u)$$

satisfies the PDE

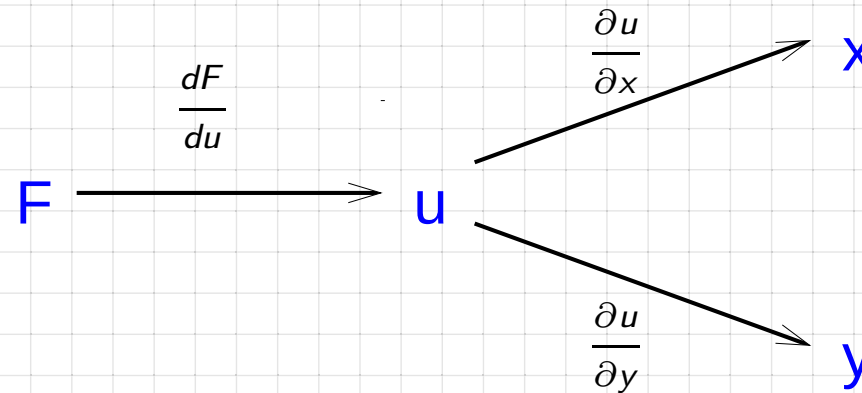
$$z_u + z_v = 0 \quad .$$

Chain Rule

If $F = F(u)$ and $u = u(x, y)$ then

$$\frac{\partial F}{\partial x} = \frac{dF}{du} \frac{\partial u}{\partial x}$$

$$\frac{\partial F}{\partial y} = \frac{dF}{du} \frac{\partial u}{\partial y}$$



To compute $\frac{\partial F}{\partial x}$ sum the paths (left to right) from F to x and multiply derivatives on a path.

Example:

Given $w = \arctan(y/x)$, calculate $\frac{\partial w}{\partial x}$ via the chain rule.

(1.6) Functions of Three (or more) Variables

If $F = F(x, y, z)$ then

$$D_1 F \equiv F_x(x, y, z) \equiv \frac{\partial F}{\partial x} = \lim_{h \rightarrow 0} \frac{F(x + h, y, z) - F(x, y, z)}{h}$$

$$D_2 F \equiv F_y(x, y, z) \equiv \frac{\partial F}{\partial y} = \lim_{h \rightarrow 0} \frac{F(x, y + h, z) - F(x, y, z)}{h}$$

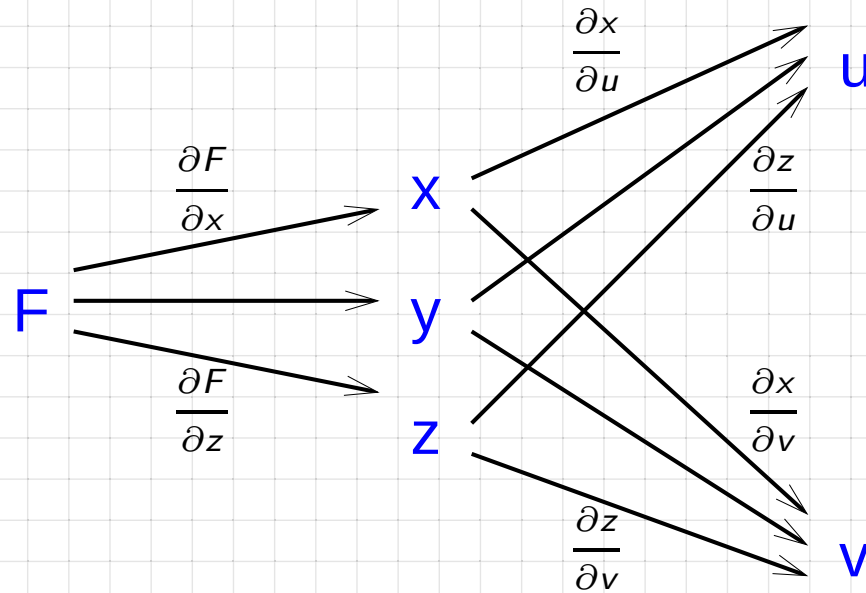
$$D_3 F \equiv F_z(x, y, z) \equiv \frac{\partial F}{\partial z} = \lim_{h \rightarrow 0} \frac{F(x, y, z + h) - F(x, y, z)}{h}$$

Differentiate w.r.t. one of the variables by holding the other variables fixed.

Chain Rule

If $F = F(x, y, z)$ and $x = x(u, v), y = y(u, v), z = z(u, v)$
then

$$\frac{\partial F}{\partial u} = \frac{\partial F}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial F}{\partial y} \frac{\partial y}{\partial u} + \frac{\partial F}{\partial z} \frac{\partial z}{\partial u}$$



Example

Find the rate of change $r(t) \equiv dw/dt$ given that

$$w(x, y, z) = xy + z$$

and the function w is sampled by the helix

$$x = \cos(t), \quad y = \sin(t), \quad \text{and} \quad z = t.$$

Example

Given $w(x, y, z) = x + 2y + z^2$ with

$x = r/s$, $y = r^2 + \ln s$, and $z = 2r$

express $\partial w / \partial r$ and $\partial w / \partial s$ in terms of r and s .

Example

Advection of concentration θ by constant velocity v is described by the PDE

$$\frac{\partial \theta}{\partial t} + v \frac{\partial \theta}{\partial x} = 0 \quad , \quad (4)$$

where the concentration $\theta(x, t)$ is the unknown function. Show that the solution to (4) is of the form $\theta(x, t) = f(x - v t)$, where f is a differentiable function.

(1.7) MAPLE NOTES

The MAPLE `plot3d` command is useful for visualization.

MAPLE also has a useful package for several variable calculus. Click on Tools in the menu bar then drag down to Tutors and across to Calculus - Multi-Variable to plot level curves and surfaces in 3D.

The MAPLE `diff` command carries out partial differentiation. For example `diff(f(x,y),x)` computes $\frac{\partial f}{\partial x}$

